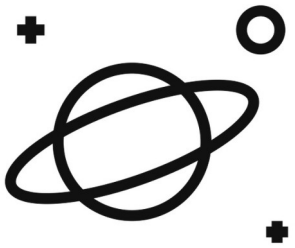
Name:

Teacher:



National 5 Physics
Space

	Success Criteria	Class	Revision	Exam
L1	I have a basic understanding of the universe			
	I can use the following terms correctly and in context: planet, dwarf planet, moon, sun, asteroid, solar system, star, exoplanet, galaxy, and universe.			
L2	I can solve problems involving weight, mass and gravitational field strength, in different locations in the universe.			
	I can describe the difference between mass and weight			
L3	I can describe several challenges of space travel.			
	I can give example of methods to maintain sufficient energy to operate life support systems in a spacecraft.			
	I can describe the risks associated with crewed space exploration.			
L4	I can describe the purpose and function of an ion drive engine			
	I can describe the purpose and function of the ‘catapult’ manoeuvre			
	I can describe the difficulties of manoeuvring a spacecraft in deep space			
L5	I can describe the benefits of satellites			
	I can state the altitude and period of orbit for a geostationary satellite			
	I can describe the relationship between altitude and period of orbit for satellites			
L6	I can state Newton's second law of motion			
	I can state Newton's third law of motion			
	I can apply Newton's second law of motion to problems involving rockets.			
	I can apply Newton's third law of motion to problems involving rockets.			
L7	I can give a basic description of the Big Bang theory			
	I can state the estimated age of the universe			
L8	I can identify continuous and line spectra.			
	I can use spectral data for known elements, to identify the elements present in stars.			
L9	I can correctly use the term light year.			
	I can convert between light years and metres.			
L10	I can describe how different parts of the electromagnetic spectrum are used to obtain information about astronomical objects.			

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Data sheet (extract)

You may use this page in every task and test in this booklet.

Gravitational field strength on the surface

Earth	Jupiter	Mars	Mercury	Moon	Neptune	Saturn	Sun	Uranus	Venus
9.8	23	3.7	4.0	1.6	11	9.0	270	8.7	8.9

All values in $N\ kg^{-1}$.

Other data

Speed of light in a vacuum: $v = 3.0 \times 10^8\ m\ s^{-1}$

One year: $1\ year = 3.15 \times 10^7\ s$

Relationships you need in this topic

$W = mg$ weight = mass × gravitational field strength

$F_u = ma$ unbalanced force = mass × acceleration

$d = vt$ distance = speed × time

Data sheet values © Qualifications Scotland (SQA).

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L1 – The solar system and beyond

- I have a basic understanding of the universe
- I can use the following terms correctly and in context: planet, dwarf planet, moon, sun, asteroid, solar system, star, exoplanet, galaxy, and universe.

Task 1:Complete the table with the SQA accepted definitions

Star	Large spherical ball of plasma, undergoing **nuclear fusion** at its core.
Planet	
Dwarf Planet	
Moon	
Asteroid	
Solar System	
Exoplanet	
Galaxy	
Universe	

Task 2: Create flashcards for the definitions

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Task 3: **In pairs, or small groups, quiz each other on the definitions**

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Task 4:Exam practice

1. The Milky Way is an example of a
   A moon
   B planet
   C solar system
   D galaxy
   E universe. (1)

2. A pupil lists five objects: the Moon, the Milky Way, the Sun, Earth, the Solar System.
   Write them in order of increasing size, smallest first. (1)

3. Astronomers have found seven rocky worlds orbiting the star TRAPPIST-1, 40 light years from Earth.
   (a) State the term used for a planet that orbits a star other than the Sun. (1)
   (b) State what is happening at the core of TRAPPIST-1 that makes it a star. (1)
   (c) State one difference between a planet and a dwarf planet. (1)

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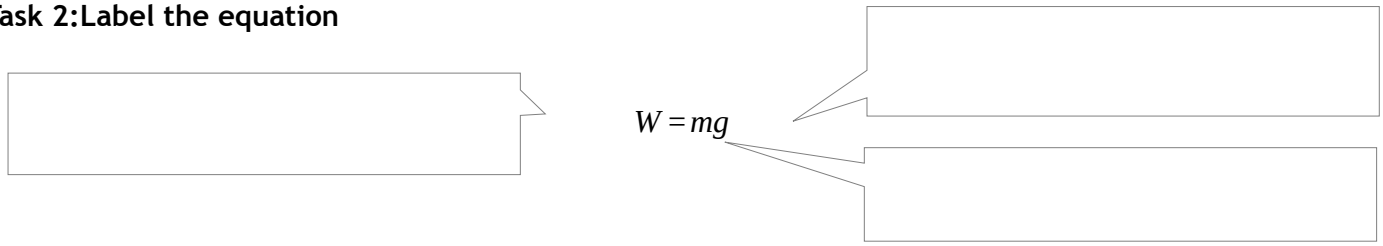
L2 – Mass and Weight

- I can describe the difference between mass and weight.
- I can solve problems involving weight, mass and gravitational field strength, in different locations in the universe.

Task 1:Complete the table with the SQA accepted definitions

Mass	The amount of _____ and object is made off. _____ _____ _____ on every planet
Weight	The _____ due to _____ of an object Is _____ on every planet

Task 2:Label the equation



Task 3:Try the example questions and get your teacher to check (data sheet and IESSUU)

1. Jimmy has a mass of 75 kg. **Calculate** Jimmy's weight on Earth.

2. Sally has a mass of 60 kg. **Calculate** Sally's weight on Mars.

3. A rock has a weight of 196 N on Earth. **Calculate the mass of the rock.**

Task 4:Exam practice

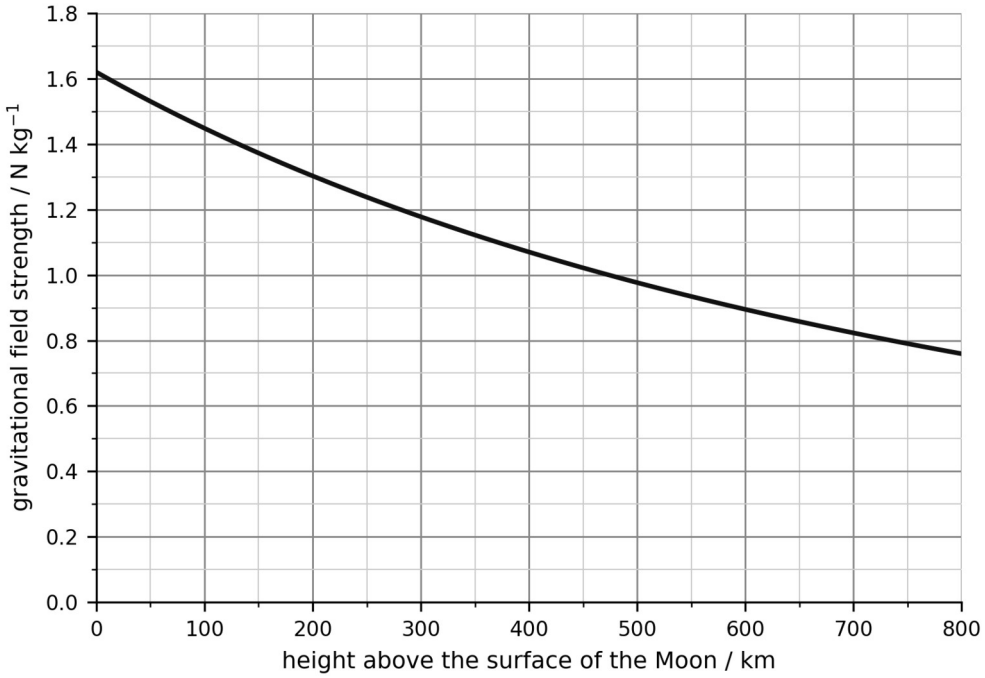
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1. A rock has a weight of 80 N on the Moon.
The gravitational field strength on the Moon is 1.6 N kg^{-1}.
The mass of the rock is

A 0 kg
B 1.6 kg
C 50 kg
D 80 kg
E 128 kg. (1)

2. A lander of mass 540 kg is sent to Mars.
Using the data sheet, calculate the weight of the lander on Mars. (3)

3. The graph shows how the gravitational field strength above the surface of the Moon changes with height.



(a) Use the graph to state the gravitational field strength at a height of 200 km. (1)
(b) A probe of mass 400 kg orbits the Moon at this height.
Calculate the weight of the probe. (3)
(c) State what happens to the mass of the probe as it climbs from the surface to 200 km. (1)

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L3 – Space Exploration and the International Space Station

- I can describe several challenges of space travel.
- I can give example of methods to maintain sufficient energy to operate life support systems in a spacecraft.
- I can describe the risks associated with crewed space exploration.

Task 1: After watching the video about life on the ISS, write down as many challenges and Solutions to surviving in space as you can remember

Challenges	Solutions

Task 2: Research the following risks of crewed space travel, then write a couple sentences describing each, and how to reduce the risk.

Risk	Description	How to reduce
Fuel load on take off		
Exposure to radiation		
Pressure differential		
Re-entry to earths atmosphere		

Task 3: Research the following solutions to maintaining energy during space travel, then write a couple sentences describing each, their benefits and drawbacks.

Solution	Description	Benefits	Drawbacks
Solar Cells			
Radioisotope Thermoelectric Generator			

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Task 4: **In small groups, design a space station that astronauts would be able to live on for at least a year.**

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Your space station must show at least three ways of overcoming the challenges of living in space, and explain the physics behind each one.

Task 5:Exam practice

1. A spacecraft sent to the outer solar system carries a radioisotope thermoelectric generator (RTG) rather than solar cells.
   The main reason for this is that
   A RTGs have a smaller mass than solar cells
   B sunlight is much weaker far from the Sun
   C RTGs produce no waste heat
   D solar cells cannot work in a vacuum
   E RTGs need no fuel of any kind. (1)

2. Astronauts on the International Space Station spend months in orbit.
   (a) State one risk to astronauts from radiation in space. (1)
   (b) Describe one method used to reduce this risk. (1)
   (c) State one way the crew is supplied with drinking water over a long mission. (1)

3. A spacecraft uses solar cells to supply energy to its life-support systems.
   (a) State one benefit of using solar cells. (1)
   (b) State one drawback of using solar cells. (1)
   (c) Name one other method of maintaining an energy supply on a long mission. (1)

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L4 – Getting around in space

- I can describe the purpose and function of an ion drive engine
- I can describe the purpose and function of the 'catapult' manoeuvre
- I can describe the difficulties of manoeuvring a spacecraft in deep space

Task 1: **On a laptop or computer go to the website https://iss-sim.spacex.com and try and dock your rocket onto the international space station.**

Task 2: **After doing task 1, use the lines below to describe why it was difficult to do.**

Task 3: **After watching the clip about the gravity assist (catapult) manoeuvre, describe it below**

Task 4:Fill in the table about types of rocket engines.

Engine	Description	Benefits	Drawbacks
Chemical Rocket			
Ion-drive			

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Extension: **Go to https://homeandcity.nasa.gov and have a look at benefits of space travel in everyday life**

Task 5:Exam practice

1. A spacecraft far from any star has its engine switched off. The forces acting on it are balanced. The spacecraft will
   - A stop immediately
   - B slow down and then stop
   - C move at a constant speed in a straight line
   - D accelerate forwards
   - E travel in a circle. (1)

2. Deep-space probes often use an ion drive engine instead of a chemical rocket engine.
   - (a) State one advantage of an ion drive engine. (1)
   - (b) State one disadvantage of an ion drive engine. (1)
   - (c) Explain why a chemical rocket engine is still needed to launch a spacecraft from the surface of the Earth. (2)

3. A space probe is steered close to Jupiter so that it gains speed.
   - (a) Name this manoeuvre. (1)
   - (b) State one benefit of using it on a long mission. (1)
   - (c) State one difficulty of steering a spacecraft in deep space. (1)

Question 1 is adapted from past paper questions. © Qualifications Scotland (SQA).

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L5 – Satellites

- I can describe the benefits of satellites
- I can state the altitude and period of orbit for a geostationary satellite
- I can describe the relationship between altitude and period of orbit for satellites

Task 1: **In the space below, write down as many benefits and uses of satellites that you know about.**

Task 2: **Watch the animation about Newton's cannon – complete the diagram below, and write a brief description of what happens underneath**



Task 3:Complete the table with the SQA accepted definitions

Geostationary Satellite	A geostationary satellite is a satellite that has an _____ period of ____ hours The orbital altitude of a geostationary satellite is _____

Task 4:Fill in the blanks below

As the _____ altitude of a satellite increases, the orbital period _____

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Extension: BBC Bitesize – Space Exploration (QR)

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Task 5:Exam practice

1. Satellite X orbits the Earth at an altitude of 400 km.
Satellite Y orbits the Earth at an altitude of 36 000 km.
Which statement is correct?
   A The orbital period of X is greater than the orbital period of Y
   B The orbital period of Y is greater than the orbital period of X
   C X and Y have the same orbital period
   D X has an orbital period of 24 hours
   E Y has an orbital period of 1 hour. (1)

2. A television company uses a satellite in a geostationary orbit.
   (a) State the orbital period of a geostationary satellite. (1)
   (b) State the approximate altitude of a geostationary orbit. (1)
   (c) State one use of satellites other than for television. (1)

3. The table shows the orbital period of four satellites at different altitudes.

Altitude / km	400	800	20 000	36 000
Orbital period	93 minutes	101 minutes	12 hours	?

   (a) Suggest a value for the orbital period of the satellite at 36 000 km. (1)
   (b) State the relationship between orbital altitude and orbital period. (1)

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L6 – Forces on Rockets

- I can state Newton's second law of motion
- I can state Newton's third law of motion
- I can apply Newton's second law of motion to problems involving rockets.
- I can apply Newton's third law of motion to problems involving rockets.

Task 1:Complete the table with the SQA accepted definitions

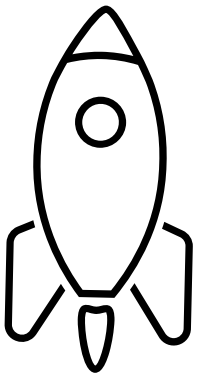
Newton's 1st Law	If the _____ on an object are balanced, then the object will move at a constant _____ in the same direction
Newton's 2nd Law	If an _____ force is applied to an object it will accelerate $F_u = ma$
Newton's 3rd Law	If object **a** exerts a _____ on object **b,** then object ___ will exert an equal but _____ force back on object ___

Task 2:Label the equation

$F_u = ma$

Task 3:Label the rocket below, showing all the forces during take off

Word bank: Thrust, Weight, Air Resistance, Unbalanced Force

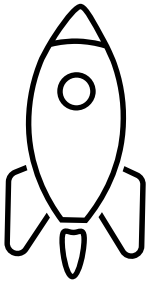


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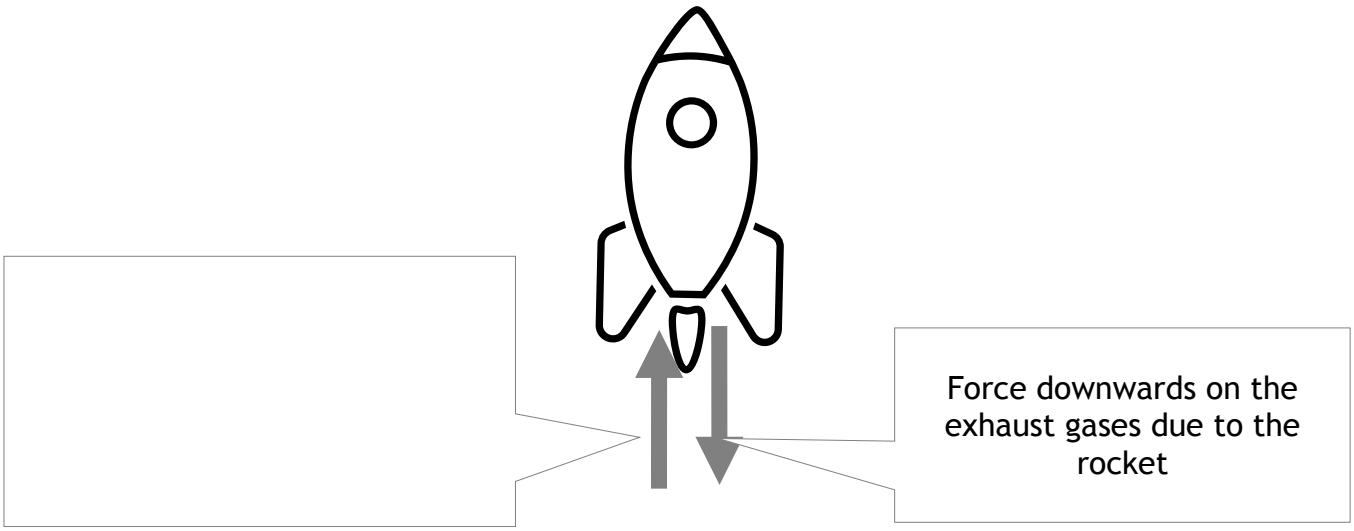
Task 4:Try the example question

A rocket has a mass of 1850 kg and the engine can produce a thrust of 30 kN

a. Complete the following diagram, showing all forces acting on the rocket.
b. Calculate the initial acceleration of the rocket.



Task 5:Complete the diagram below, to show the Newton pair acting on the rocket.



Task 6:Exam practice

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1. A rocket engine pushes exhaust gases downwards with a force of 500 000 N.
The force exerted on the rocket by the exhaust gases is

A 0 N
B 250 000 N downwards
C 500 000 N downwards
D 500 000 N upwards
E 1 000 000 N upwards. (1)

2. A rocket has a mass of 12 000 kg. Its engine produces a thrust of 150 000 N.
Air resistance can be ignored.

(a) Calculate the weight of the rocket on Earth. (3)
(b) Calculate the unbalanced force on the rocket at lift-off. (1)
(c) Calculate the initial acceleration of the rocket. (3)

3. A probe in deep space fires its engine for 10 seconds and then switches it off.

(a) State Newton's third law of motion. (1)
(b) Describe the motion of the probe after the engine is switched off. (1)

Question 1 is adapted from past paper questions. © Qualifications Scotland (SQA).

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Review: Check your booklet is up to date
Test yourself on the definitions
Work through the Space practice questions on the website (Parts A and B)
Check your answers against the mark scheme on the page
Fill in your learning Log

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L7 - The Big Bang Theory

- I can give a basic description of the Big Bang theory
- I can state the estimated age of the universe

Task 1:Complete the table with the SQA accepted definitions

Age of the universe	The universe is approximately_____ _____ years old

Task 2: After watching the clip about the Big Bang theory, describe it below

Extension: Research each of the following pieces of evidence for the Big Bang theory and briefly describe them below.

Evidence	Description
Cosmic Microwave Background Radiation	
Olbers' paradox	
The abundance of light elements in space	

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Task 3:Exam practice

1. The estimated age of the universe is
   - A 4.6 thousand years
   - B 4.6 million years
   - C 13.8 million years
   - D 13.8 billion years
   - E 13.8 trillion years. (1)

2. The Big Bang theory describes how the universe began.
   - (a) Describe what the Big Bang theory says about how the universe began. (2)
   - (b) State the estimated age of the universe. (1)

3. Cosmic microwave background radiation is detected coming from every direction in space.
   - (a) State what this observation is evidence for. (1)
   - (b) Name one other piece of evidence that supports the Big Bang theory. (1)

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L8 – Spectra

- I can identify continuous and line spectra.
- I can use spectral data for known elements, to identify the elements present in stars.

Task 1: **Using the spectra data collected in your experiment, determine the elements present in the mystery sample.**
Use coloured pencils, and the spaces below to record each individual spectrum.

Element	Spectra
Mix	

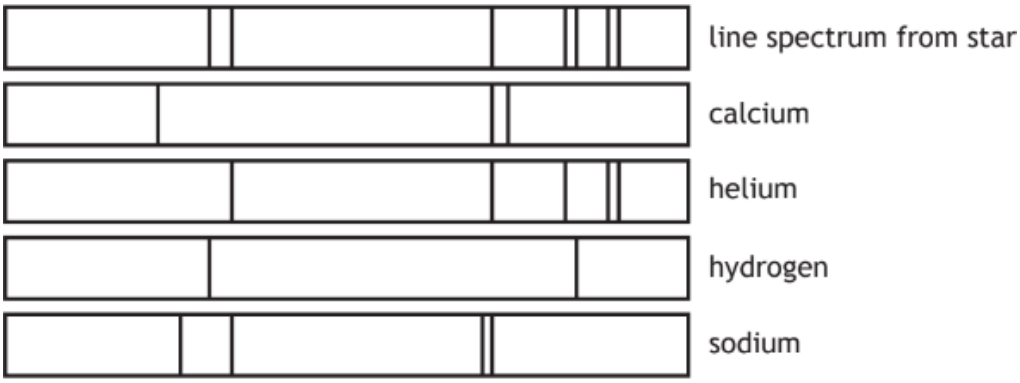
The mixture contains: _____

Task 2:Using colour pencils, use the spaces below to show both types of spectra.

Continuous	
Line	

Task 3:Try the example question:

Using the spectral data below, determine what element(s) are present in the star bellow.

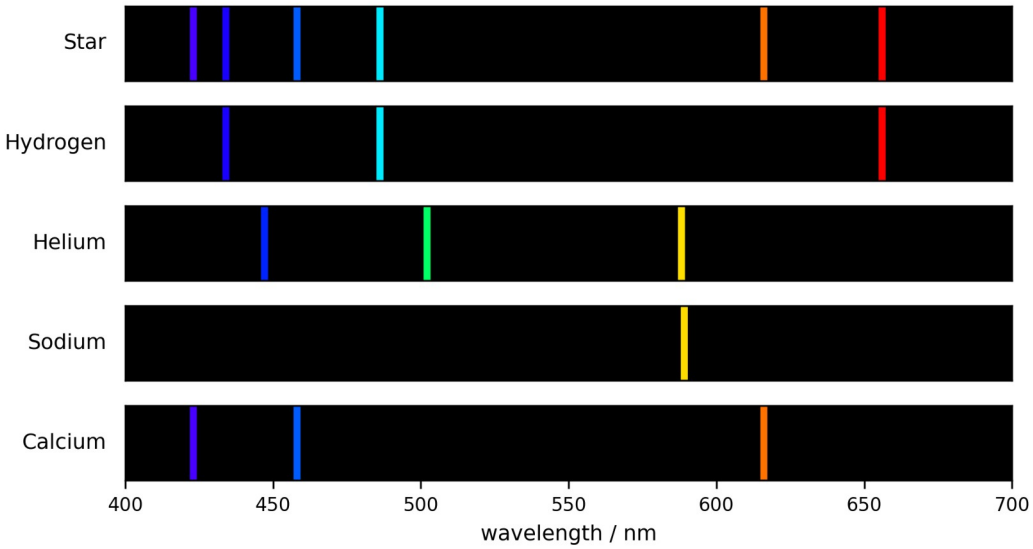


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Task 4:Exam practice

1. The spectrum from a filament lamp shows all the colours of the visible spectrum with no gaps. This is an example of
   - A a line spectrum
   - B an absorption spectrum
   - C a continuous spectrum
   - D a radio spectrum
   - E a gamma spectrum. (1)

2. The spectrum of a star is shown, together with the line spectra of four known elements.



   (a) State the type of spectrum produced by the star. (1)
   (b) Identify the elements present in the star. (2)

3. Astronomers study starlight to work out what stars are made of.
   (a) Name the instrument used to observe the spectrum of light from a star. (1)
   (b) Explain how astronomers can tell which elements are present in a star. (1)

Question 2 is adapted from past paper questions. © Qualifications Scotland (SQA).

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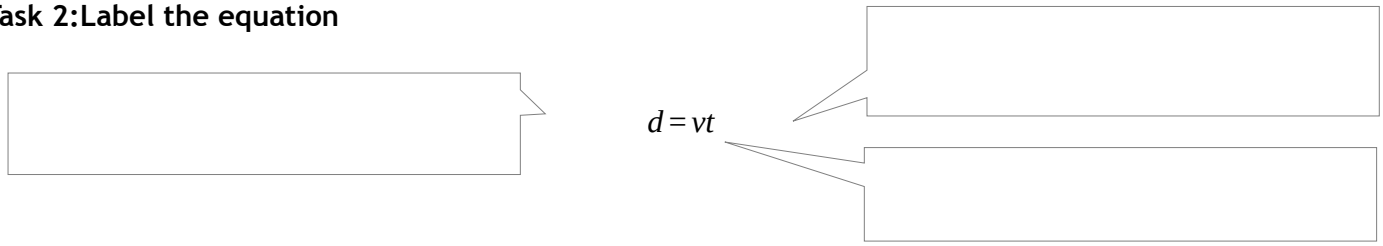
L9 – Light Years

- I can correctly use the term light year.
- I can convert between light years and metres.

Task 1:Complete the table with the SQA accepted definitions

Light year	The **distance** travelled by light in_____ year

Task 2:Label the equation



Examples: It is very important you lay these out in the same way during tests.

A spacecraft is 1 light year away. Calculate the distance in metres (use your data sheet to help)

$v = 3 \times 10^8$ $d = ?$ $t = 1 \text{ year} = (1 \times 365 \times 24 \times 60 \times 60)\text{ s}$	$d = vt$ $d = 3 \times 10^8 \times (1 \times 365 \times 24 \times 60 \times 60)$ $d = 9.46 \times 10^{15} m$

The star Proxima Centauri is 4.2 light years away. Calculate the distance in metres (use your data sheet to help)

$v = 3 \times 10^8$ $d = ?$ $t = 4.2 \text{ year} = (4.2 \times 365 \times 24 \times 60 \times 60)\text{ s}$	$d = vt$ $d = 3 \times 10^8 \times (4.2 \times 365 \times 24 \times 60 \times 60)$ $d = 3.97 \times 10^{16} m$

Task 3: Exam practice

1. One light year is
   - A the time light takes to travel for one year
   - B the distance light travels in one year
   - C the distance from the Earth to the Sun
   - D the speed of light over one year
   - E the time taken for light from the Sun to reach the Earth. (1)

2. The star Sirius is 8.6 light years from Earth.
   Calculate this distance in metres. (4)
   Take 1 year to be 3.15×10^7 s.

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3. Light from a distant galaxy takes 2.5 million years to reach Earth.

(a) State the distance to the galaxy in light years. (1)

(b) Explain why astronomers measure distances in space in light years rather than in metres. (1)

(c) State what astronomers are seeing when they look at this galaxy today. (1)

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Extension: BBC Bitesize Cosmology **(QR)**

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L10 - Telescopes

- I can describe how different parts of the electromagnetic spectrum are used to obtain information about astronomical objects.

Task 1: Fill in the blanks below:

The electromagnetic spectrum						
Longest Wavelength						Shortest Wavelength
Radio Waves						

Task 2: Answer The Questions

1. What speed do microwaves travel at? _____
2. Why do telescopes use different parts of the EM Spectrum? _____

   _____
3. What part of the EM Spectrum does the Hubble telescope (mainly) use? _____
4. What part of the EM Spectrum does the James Webb telescope (mainly) use? _____
5. How do the pictures of 'the pillars of creation' compare for each of these? _____

   _____

   _____

Task 3: Go to the NASA website (QR), Pick one of the recent James Webb discoveries and write a paragraph below to summarise what you have learned

_____

_____

_____

_____

_____

_____

_____

Task 4:Exam practice

1. Which part of the electromagnetic spectrum has the longest wavelength?
   A gamma rays
   B X-rays

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C ultraviolet
D infrared
E radio waves. (1)

2. The James Webb Space Telescope detects infrared radiation.
   (a) State the speed at which infrared radiation travels through space. (1)
   (b) Explain why an infrared telescope can see through clouds of dust that block visible light. (1)
   (c) Name one part of the electromagnetic spectrum, other than infrared and visible light, used to study astronomical objects. (1)

3. Some telescopes are placed in orbit above the Earth's atmosphere.
   (a) State one advantage of placing a telescope in orbit. (1)
   (b) State one part of the electromagnetic spectrum that can be detected by telescopes on the Earth's surface. (1)

Review: Check your booklet is up to date
Test yourself on the definitions
Sit Space Practice Test A on the website
Then sit Practice Test B and compare your scores
Fill in your learning Log
Revise for the Space Test

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